

Short Presentation Note: Markov Locality, Diffusion Fill-In, and the First Non-Gaussian Route

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Abstract

The Gaussian AR(1) benchmark is solved. What remains: how local Markov structure appears in the score, how diffusion reshapes it, how noise erases it. This note presents the strongest benchmark facts, a deterministic control experiment that separates propagation-induced couplings from destroyed Markov locality, and the first non-Gaussian extension (Laplace) with exact Fourier representation and position-dependent score Hessian. Figures are primary; derivations are in the full manuscript.

1 Benchmark and Open Problem

Consider the AR(1) model:

$$a_{k+1} = \alpha a_k + \eta_k, \quad \eta_k \sim \mathcal{N}(0, \sigma_\eta^2), \quad |\alpha| < 1,$$

with prior $a_0 \sim \mathcal{N}(\mu_0, \Sigma_0)$. Under Ornstein–Uhlenbeck noising, the diffused state at time t is:

$$x_k = e^{-t} a_k + \sqrt{\Delta_t} z_k, \quad \Delta_t = 1 - e^{-2t},$$

with marginal covariance:

$$\Sigma_t = e^{-2t} \Sigma_0 + \Delta_t \mathbf{I}.$$

For Gaussian $p_t(x)$, the score is affine:

$$S(x, t) = \nabla \log p_t(x) = -Q_t(x - \mu_t),$$

where $Q_t = \Sigma_t^{-1}$ is the *precision matrix* and $\gamma_t = e^{-2t}/\Delta_t$.

Two organizing ingredients:

- (i) **Gaussianity:** Affine score, spectral decoupling in eigenbasis.
- (ii) **Markovianity:** AR(1) structure makes Q_0 *tridiagonal*, encoding only nearest-neighbor interactions.

Central question: How does this local Markov structure survive, deform, and disappear in the score under diffusion?

2 Gaussian Benchmark: The Precision Lifecycle

Structure and Spectral Dynamics

The initial precision is tridiagonal:

$$Q_0 = \begin{pmatrix} q_0 & -q_1 & 0 & \cdots \\ -q_1 & q_0 & -q_1 & \cdots \\ 0 & -q_1 & q_0 & \cdots \\ \vdots & \vdots & \vdots & \ddots \end{pmatrix}$$

with all structure concentrated in the first three diagonals. Covariance contracts uniformly:

$$(\Sigma_t)_{jk} = e^{-2t}(\Sigma_0)_{jk} \quad \text{for } j \neq k.$$

The precision restructures via spectral decomposition:

$$Q_t = U \operatorname{diag} \left(\frac{1}{e^{-2t}\lambda_i + \Delta_t} \right) U^\top = \frac{1}{\Delta_t} (I + \gamma_t \Sigma_0)^{-1}.$$

Coupling distance scales at small times: Distance- d couplings appear at order t^{d-1} ,

$$Q_t = e^{2t} \sum_{m \geq 0} (-c_t)^m Q_0^{m+1}, \quad c_t = e^{2t} - 1.$$

This geometric series encodes band-by-band fill-in: first neighbors at $O(1)$, second at $O(t)$, third at $O(t^2)$, etc.

Large-time asymptotics:

$$Q_t \approx I - e^{-2t}(\Sigma_0 - I).$$

The precision becomes nearly identity as diffusion dominates.

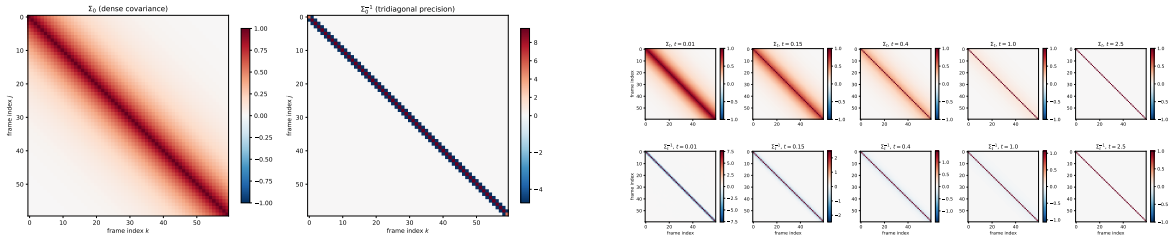


Figure 1: **Left:** Initial covariance Σ_0 (dense, rank-full) vs. initial precision Q_0 (tridiagonal, Markov). **Right:** Precision lifecycle: band-by-band fill-in and diffusion-driven relaxation toward identity.

3 Locality Diagnostics

To quantify how Markov structure persists or dissolves, we define four observables:

1. **Tridiagonal mass:** $T_1(t) = \sum_{i,j:|i-j|\leq 1} Q_t[i,j]^2 / \|Q_t\|_F^2$.
2. **Effective bandwidth:** $b_{0.95}(t) = \min\{d : \sum_{|i-j|\leq d} Q_t[i,j]^2 \geq 0.95 \|Q_t\|_F^2\}$.
3. **Interaction range:** $\bar{\xi}(t) = \sum_{d \geq 1} d \cdot P(|i-j| = d)$ where P is the band-mass distribution.

4. **Neighbor score correlation:** $\rho_1(t) = \text{Corr}(S_k, S_{k+1})$, read directly from Q_t .

Since $\text{Cov}(S_t) = Q_t$ (the precision itself), these observables fully characterize locality in the score. The posterior mean is:

$$\mathbb{E}[a \mid X_t = x] = \mu_0 + G_t(x - \mu_t), \quad G_t = e^{-t\Sigma_0}Q_t.$$

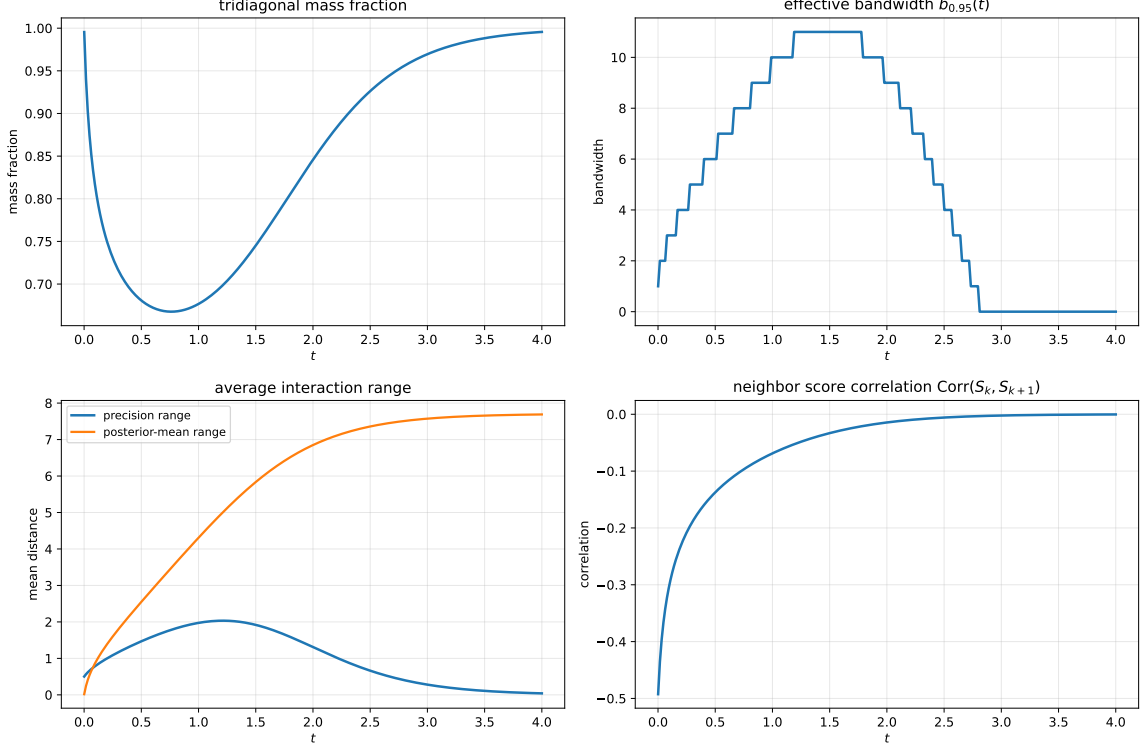


Figure 2: **Four-panel locality diagnostics:** Tridiagonal mass fraction $T_1(t)$ decays; bandwidth $b_{0.95}(t)$ grows; interaction range $\bar{\xi}(t)$ increases; neighbor correlation $\rho_1(t)$ vanishes. Diffusion simultaneously fills the precision and decorrelates it.

4 Deterministic Control Experiment

To isolate how Markov structure is destroyed by diffusion, we study a deterministic analogue with no stochastic innovation noise:

$$a_k = \alpha^k A_0, \quad \Sigma_0^{\text{det}} = \sigma_0^2 \mathbf{v} \mathbf{v}^\top, \quad \mathbf{v} = (1, \alpha, \dots, \alpha^K)^\top.$$

By Sherman–Morrison, after noising:

$$Q_t^{\text{det}} = \frac{1}{\Delta_t} I - C(t) \mathbf{v} \mathbf{v}^\top.$$

Key observation: The precision becomes *globally dense immediately*, even for arbitrarily small $t > 0$. The rank-one structure of Σ_0^{det} induces global couplings purely through *propagation*, independent of diffusion destroying Markov locality.

Noncommuting limits:

- $(t \rightarrow 0, \sigma_\eta^2 > 0)$: Tridiagonal Q_0 .
- $(\sigma_\eta^2 = 0, t > 0)$: Rank-one-modified identity, globally dense.

This experiment cleanly separates *diffusion-destroyed Markov locality* from *propagation-induced global coupling*.

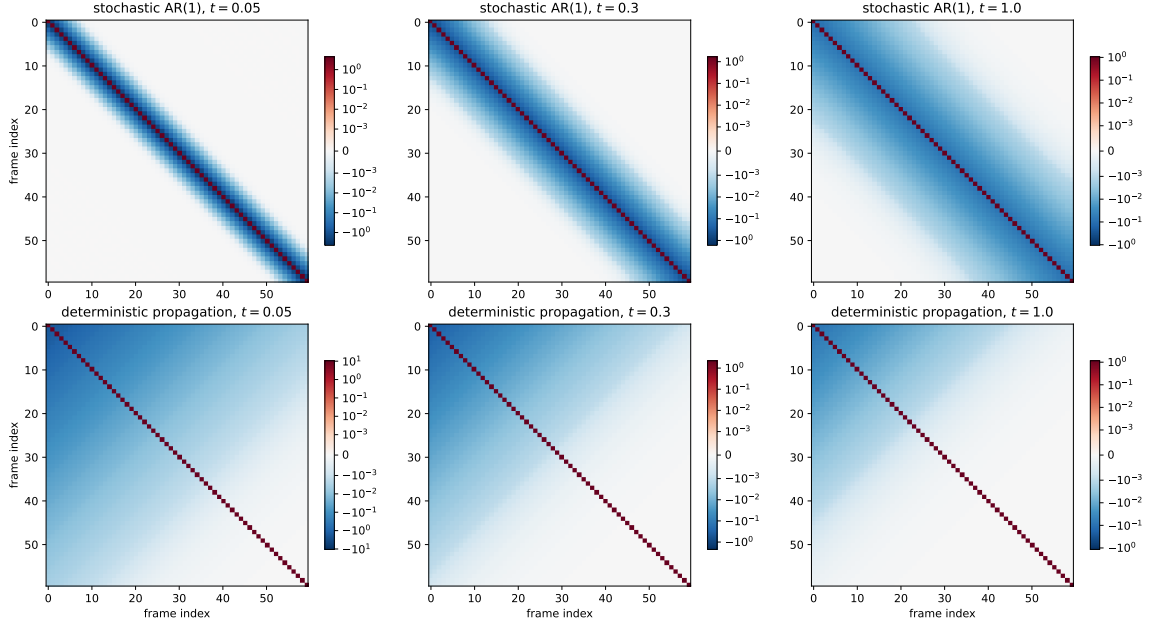


Figure 3: **Stochastic (left) vs. deterministic (right) precision matrices:** Stochastic AR(1) shows band-by-band fill-in and eventual diffusion-driven relaxation. Deterministic model produces global coupling from propagation alone, illustrating that the source of global structure can be separated from diffusion destruction of local Markov form.

5 First Non-Gaussian Extension: Laplace

The Gaussian benchmark is solved. We now take the first step beyond Gaussianity.

Model and Fourier Representation

Let $a_{k+1} = \alpha a_k + \varepsilon_k$ with $\varepsilon_k \sim \text{Laplace}(0, b)$. The characteristic function is $\hat{f}(q) = \frac{1}{1+b^2 q^2}$, making $\text{Laplace}(0, b)$ the low-frequency tangent to $\mathcal{N}(0, 2b^2)$.

Under diffusion, the exact Fourier representation is:

$$\Phi_t(\xi) = \exp\left(-\frac{\Delta_t \|\xi\|^2}{2}\right) \cdot \Phi_0(e^{-t}\xi).$$

For $K = 1$ (scalar a_0), the posterior density after noising is computed via Laplace–Gaussian convolution:

$$h(y; \tilde{b}, \sigma^2) = \frac{1}{4\tilde{b}} \exp\left(\frac{\sigma^2}{2\tilde{b}^2}\right) \left[\exp(y/\tilde{b}) \operatorname{erfc}(\cdots) + \exp(-y/\tilde{b}) \operatorname{erfc}(\cdots) \right],$$

followed by 1D Gaussian integration over a_0 . The exact formula is closed-form, validated against Monte Carlo.

Score and Position-Dependent Hessian

The score $S(x, t) = \nabla \log p_t(x)$ is now **nonlinear**. Critically, the score Hessian:

$$H_t(x) = -\nabla^2 \log p_t(x)$$

depends on x . In the Gaussian case, $H_t \equiv Q_t$ is constant. Here, the "effective precision" varies with location in data space.

For $K = 2$, the Markov chain structure is preserved: the density factors as a chain of 1D integrals, each solvable by the same closed-form Laplace–Gaussian recipe.

Monte Carlo validation: Numerical integration confirms the exact computation to machine precision.

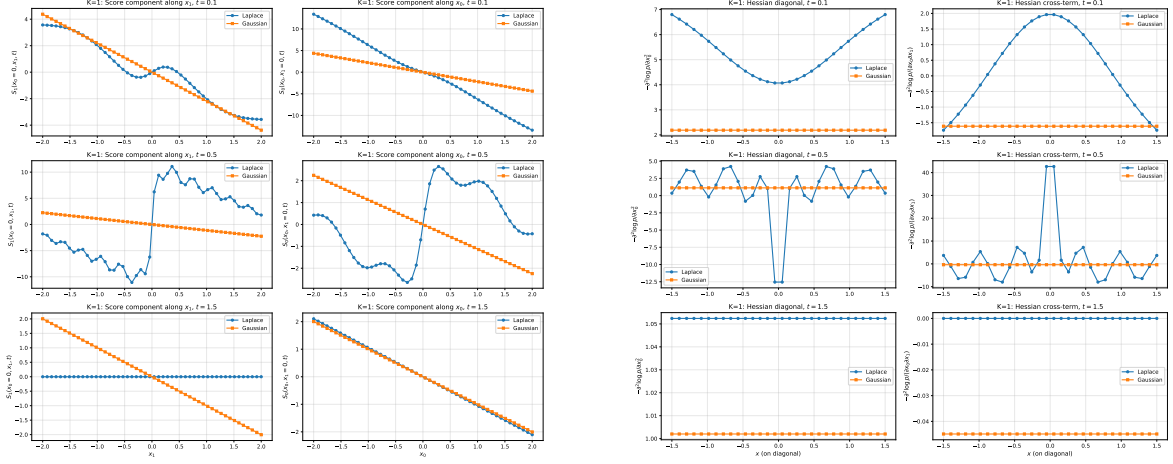


Figure 4: **Left:** Score in Laplace model (blue, nonlinear) vs. Gaussian (orange, affine) for $K = 1$. **Right:** Score Hessian $H_t(x)$ (effective precision) is position-dependent in Laplace; constant identity-like in Gaussian (dashed). This x -dependence is the hallmark of non-Gaussian score structure.

6 Takeaways and Next Steps

1. **Markov structure lives in precision, not covariance:** Q_0 is tridiagonal while Σ_0 is dense.
2. **Band-by-band fill-in:** Distance- d couplings enter at order t^{d-1} . Small-time expansion encodes the geometric series $Q_t = e^{2t} \sum_{m \geq 0} (-c_t)^m Q_0^{m+1}$.
3. **Intermediate regime captured by locality observables:** Tridiagonal mass, band-width, interaction range, and neighbor correlation quantify the transition from local to global structure.
4. **Deterministic benchmark separates mechanisms:** Global couplings can arise from propagation alone; diffusion’s destruction of Markov locality is a distinct effect.
5. **Laplace model: first exact non-Gaussian extension:** Fourier representation, closed-form 1D integrals, and validated numerics enable systematic non-Gaussian analysis.
6. **Score Hessian becomes position-dependent:** In Laplace, $H_t(x) \neq \text{const}$, replacing the global effective precision of the Gaussian case with a local, data-dependent curvature.
7. **Next direction:** Systematically study band mass of $H_t(x)$ as a function of (x, t) and extend to higher-order non-Gaussian models (e.g., Student- t , mixture innovations).

Key insight: The Gaussian AR(1) diffusion score is solved. The first non-Gaussian extension shows that locality becomes both space-dependent and model-dependent. This opens the door to understanding how diffusion reshapes and erases Markov structure in realistic, non-Gaussian settings.

References

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